

Subject Description Form

Subject Code	EEE509
Subject Title	Micro- and Nano-photonic Semiconductor Devices
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	To study an advanced theoretical framework and Micro- and Nano-photonic sub-systems from the device cell structures to the array. Learn the operation principles of a large set of photonic devices. And develop sufficient knowledge with which the students can later learn about newly developed devices for future photonic applications.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) Receive a deep understanding of the mathematical and quantum-mechanical tools to design optoelectronic devices. (b) Receive an understanding of the operation principles of a large set of photonic devices such as detectors, light sources, modulators, and others. (c) Understand the operation of micro and nanophotonic semiconductor devices. (d) Develop sufficient knowledge with which the students can later learn about newly developed devices for future photonic applications.
Subject Synopsis/ Indicative Syllabus	Physics of semiconductors: Basic properties of semiconductors, comparative study of the whole set of semiconductors. Phonons: Optical, acoustical; transverse, longitudinal. Sources: LED, Lasers (Gain, non-parabolic effects, strain effects). Detectors: PIN, Avalanche, SiGe, Infrared, Metal-Schottky, Quantum Well IR, Quantum Dot IR, Thermal, Seebeck detectors. Modulators: Electro-absorption, quantum confined stark effect, electro-optic modulation. Advanced photonic semiconductor components: Quantum dots, wires, quantum cascade lasers Outlook: Recent progress and future development in semiconductor devices and their integration.

Teaching/Learning Methodology	Lecture: Basic knowledge of quantum physics and semiconductor physics will be explained. Concrete examples will be given to the design of basic semiconductor components. Assignment sets will be given to assess the learning progress of students. Tutorial: Students will work on problem sets in the tutorials, which provide them opportunities to apply the knowledge gained in lectures.																																	
	Teaching/Learning Methodology		Intended subject learning outcomes																															
			a	b	c	d																												
	Lectures		√	√	√	√																												
	Tutorials		√	√	√	√																												
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="4">Intended subject learning outcomes to be assessed (Please tick as appropriate)</td></tr><tr><td>a</td><td>b</td><td>c</td><td>d</td></tr><tr><td>1. Assessments</td><td>40%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Examination</td><td>60%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100 %</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Assignments: Assignments are given to students to assess their competence level of knowledge and comprehension. The criteria (i.e., what to be demonstrated) and level (i.e., the extent) of achievement will be graded. These will be made known to the students before an assignment is given. Feedback about their performance will be given promptly to students to help them to improve their learning. End-of-semester Examination: There will be an end-of-semester examination to assess students’ achievement of all the learning outcomes.						Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				a	b	c	d	1. Assessments	40%	√	√	√	√	2. Examination	60%	√	√	√	√	Total	100 %				
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		a	b	c	d																													
1. Assessments	40%	√	√	√	√																													
2. Examination	60%	√	√	√	√																													
Total	100 %																																	
Student Study Effort Expected	Class contact:																																	
	▪ Lecture			33 Hrs.																														
	▪ Tutorial			6 Hrs.																														
	Other student study effort:																																	
	▪ Reading and self-study			81 Hrs.																														
	Total student study effort			120 Hrs.																														

Reading List and References	Textbook • S.M. Sze, Kwok K. Ng, Physics of Semiconductor Devices, John Wiley & Sons, 2006. References • Chuang S L, Peyghambarian N, Koch S. Physics of optoelectronic devices. 1996. • Piprek J. Semiconductor optoelectronic devices: introduction to physics and simulation. Elsevier, 2013.
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